

Title: 气体动理论

Date: / /

$$n = \frac{N}{V} \quad \text{mol} \cdot \text{K}$$

$$PV = nRT \quad R = 8.31 \text{ J/mol} \cdot \text{K}$$

$$P = nKT \quad K = 1.38 \times 10^{-23} \text{ J/K}$$

$$n = \frac{N}{V}$$

$$\textcircled{1} PV = nRT = \frac{m}{M} RT$$

$$\textcircled{2} P = nKT \quad (n = \frac{N}{V})$$

$$K = 1.38 \times 10^{-23} \text{ J/K}$$

$$T = 273 + t \quad \text{热力学温标}$$

$$1 \text{ atm} = 1.01325 \text{ Pa}$$

$$P = 1 \text{ atm} \quad T = 273 \text{ K}$$

$$P = \frac{N}{V} KT$$

$$1.38 \times 10^{-23} = N$$

$$3 \times 10^{-23} \times 1.38$$

$$10^{-23}$$

考点	重要程度	占 分	常见题型
1. 状态方程	★★★★★	2~4	选择、填空
2. 内能	★★★★★	2~4	

1. 状态方程

理想气体分子模型

- (1) 忽略分子内部的结构和大小;
- (2) 除碰撞瞬间外, 分子间的相互作用力可忽略不计;
- (3) 分子间及分子与器壁的碰撞是完全弹性的。

平衡态

系统不受外界影响的条件下, 系统各部分的宏观性质长时间内不发生变化

准静态过程(平衡过程)

如果过程进展十分缓慢, 使所经历的一系列中间状态都无限接近平衡态

理想气体状态方程:

$$\textcircled{1} PV = \frac{m}{M} RT$$

$$R = 8.31 \text{ J/(mol} \cdot \text{K)}$$

$$\textcircled{2} P = nKT$$

$$\textcircled{3} P = nkT$$

$$\textcircled{4} P = nkT$$

$$\textcircled{5} P = nkT$$

$$\textcircled{6} P = nkT$$

$$\textcircled{7} P = nkT$$

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$$\textcircled{65} P = nkT$$

蜂考 bilibili

2. 内能

分子内能 = 分子平动动能 + 分子转动动能

蜂考 bilibili

分子类型

自由度

平动动能

转动动能

总动能(内能)



$$i=3$$

$$\bar{\epsilon} = \frac{3}{2} kT$$

$$\bar{\epsilon} = 0$$

$$\bar{\epsilon} = \frac{3}{2} kT$$



$$i=5$$

$$\bar{\epsilon} = \frac{5}{2} kT$$

$$\bar{\epsilon} = \frac{2}{2} kT$$

$$\bar{\epsilon} = \frac{5}{2} kT$$



$$i=6$$

$$\bar{\epsilon} = \frac{6}{2} kT$$

$$\bar{\epsilon} = \frac{3}{2} kT$$

$$\bar{\epsilon} = \frac{6}{2} kT$$

能量按自由度均分定理: $\bar{\epsilon} = \frac{1}{2} kT$

$$K \text{ 气体分子 } K = 1.38 \times 10^{-23} \text{ J/K}$$

$$\text{平动动能: } \bar{\epsilon} = \frac{3}{2} kT$$

$$\text{内能: } \bar{\epsilon} = \frac{i}{2} kT$$

$$\text{内能: } E = \frac{i}{2} \frac{m}{M} RT$$

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考点	重要程度	占 分	常见题型
1. 速度分布函数	★★★	2~10	选择、填空为主, 偶尔大题
2. 三种常用速率	★★★	0~2	

1. 速度分布函数

麦克斯韦分布函数

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-\frac{mv^2}{2kT}}$$

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1. 速度分布函数

题3. 有N个假想的气体分子, 其速率分布如图所示(当 $v > 2v_0$ 时, 分子数为零) 试求:

(1) 写出分布函数 $f(v)$ 的表达式

(2) 由N和 v_0 求a值

(3) 速率在 $1.5v_0 \sim 2v_0$ 的分子数

(4) 分子的平均速率和方均根速率

解: (1) $f(v) = \frac{N}{N} = 1$

(2) $f(v) = \frac{N}{N} = 1$

(3) $f(v) = \frac{N}{N} = 1$

(4) $f(v) = \frac{N}{N} = 1$

(5) $f(v) = \frac{N}{N} = 1$

(6) $f(v) = \frac{N}{N} = 1$

(7) $f(v) = \frac{N}{N} = 1$

(8) $f(v) = \frac{N}{N} = 1$

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(12) $f(v) = \frac{N}{N} = 1$

(13) $f(v) = \frac{N}{N} = 1$

(14) $f(v) = \frac{N}{N} = 1$

(15) $f(v) = \frac{N}{N} = 1$

(16) $f(v) = \frac{N}{N} = 1$

(17) <